

Ciaran Nimick

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Professional Summary:

Highly motivated Mechanical Engineering student seeking a Summer 2026 full-time engineering internship in aerospace, defense, or advanced manufacturing. Proven ability in mechanical design, Finite Element Analysis (FEA), and experimental validation using tools like Siemens NX, Ansys, and SolidWorks, gained through motorsports and research projects

Education:

Case Western Reserve University - Cleveland, OH

Expected Spring 2028

Bachelor of Science in Mechanical Engineering | Minor in Supply Chain Management | GPA: 3.4

Relevant Coursework: Thermodynamics | Statics and Strengths of Materials | Mechanical Manufacturing

Experience:

CWRU Motorsports Baja SAE, Cleveland OH

September 2024 - Present

- Operated manual machining tools, including lathe, mill, and tube bender, to fabricate over 15 unique steel, aluminum, and bronze components with tolerances as tight as .001 inches.

Suspension Engineer

September 2024 - Present

- Currently designing the front lower suspension arm, projecting a 10% increase in ride clearance and a 25% weight reduction while improving serviceability.
- Executed Finite Element Analysis (Ansys) on front outboard components to validate structural integrity under maximum dynamic loads, ensuring a factor of safety (FoS) of ~1.5 for all critical parts.
- Aided in developing vehicle setups for different racing environments through adjustments to suspension and steering.
- Developed a Java-based design tool that produces a chart of feasible tie rod dimensions and graphically visualizes the factor of safety.

Test Engineer

September 2024 - Present

- Analyzed and designed a custom adapter to reliably transmit ~500 ft lbs of torque for critical Continuously Variable Transmission (CVT) development.
- Designed a CVT Dyno shroud in Siemens NX engineered to dissipate 20kW of heat and maintain safe operating temperatures during extensive testing.

Case Western Reserve University, Cleveland OH - Research Assistant

March 2025 - Present

- Partnered with graduate researchers to explore magnetoreception in animals, aiming to translate biological navigation strategies into tools for interplanetary exploration.
- Aided in developing a predictive model for satellite movement in extraterrestrial magnetic fields, incorporating generalizable planetary and orbital parameters.

Case Western Reserve University, Cleveland, OH - Mechanical Manufacturing TA

August 2025 - Present

- Instructed and mentored 16 students in DFMA principles and SolidWorks part design, assemblies, and drawings, preparing them for the Certified SOLIDWORKS Associate exam.
- Graded ~180 manual sketching assignments, ensuring parts were fully defined and correct projections were drawn.

PVPHS VEX Robotics, Palos Verdes CA - President, Project Manager, & Lead Design

August 2020- June 2024

- Increased annual funding fourfold, tripled team membership, and doubled the amount of competition teams.
 - Organized an inaugural robotics event, raising \$8,000, earning congressional recognition from Congressman Ted Lieu
 - Led the design and prototyping of twelve competition robots using Onshape and Fusion 360 to implement team solutions.
 - Won five local and state awards, leading to three State Championship and two National Championship qualifications.
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Skills:

Design: Siemens NX | SolidWorks | Onshape | Fusion 360 | Ansys | GD&T

Manufacturing: Mill | Lathe | Waterjet | CNC Router | Laser Cutter | Drill Press | Band Saw | Tube Bender

Software Languages: MatLab | C++ | Java